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# Regenerative Therapy Modality for Treatment of True Combined Endodontic-Periodontal Lesions: A Randomized Controlled Clinical Trial

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## 1. Introduction

Preserving the natural dentition is the goal of every dental treatment. Periodontal treatment not only poses the challenge of preserving the natural teeth, but also of restoring the lost periodontal tissue [1]. Lesions of the periodontal ligament and adjacent alveolar bone may originate from infections of the periodontium or tissues of dental pulp. The nature of the pathological communication between the pulp and the periodontium in true combined endodontic-periodontal lesions (EPL) present a dilemma for the clinician in the diagnosis, treatment and prediction of the prognosis of the treatment [2]. The clinical conditions of EPL present with chronic progression in subjects with periodontitis, or in acute form in teeth associated with traumatic injuries or iatrogenic event, with common signs and symptoms of negative or altered pulp response to vitality tests, deep periodontal pockets, bone resorption, purulent exudate, spontaneous pain and tooth mobility [3,4]. EPL have received several classifications based on the primary source of infection [5]. A recent classification for EPL was revised by Herrera et al. in 2017 and proposed based on the present disease status and on the prognosis of the involved tooth [6].

Treatment of EPL should be in an integrated manner with both endodontic therapy and periodontal regenerative procedure to ensure a successful treatment outcome [7]. Evidence has showed that endodontic infection has a negative impact on periodontal healing, therefore, endodontic therapy should be performed first [8,9]. Currently, the root canal system is filled with gutta-percha-based materials (GP) in combination with endodontic sealer in EPL [10]. With advancements in new endodontic materials, mineral trioxide aggregate (MTA) might become a viable alternative treatment option in endodontic treatment of combined EPL. Several clinical studies have showed that the success rate is greater when using MTA as root-end filling in endodontic surgery compared to other materials or with smoothing of the orthograde GP root filling [11–14]. MTA exhibits superior sealability against bacterial micro leakage, while demonstrating antibacterial and bio-inductive properties that promotes biologic repair and regeneration of periodontal ligaments (PDL) [13,14]. Hence, the use of MTA as an obturation material might ultimately provide long-term benefits that enhance the prognosis and retention of the involved teeth with EPL [15].

True combined EPL has a more compromised prognosis than the other types of endodontic-periodontal diseases and this depends on the severity of periodontal disease and the response to periodontal treatment [5,16]. Kim et al. report that only 17 out of 42 cases of true combined EPL showed complete healing with the reestablishment of the lamina dura at 2 years recall visits [14]. A significant number of diverse treatment approaches were used to regenerate periodontal tissues including guided tissue regeneration (GTR) using barrier membranes, various types or a combination of grafting materials, enamel matrix proteins, and autologous platelet concentrates [17,18]. Cortellini et al. showed that 92% of severely compromised teeth with attachment loss to the apex successfully regenerated after periodontal regeneration at 5-year examination visit [19]. Another clinical study by Oh et al. showed that periodontal regenerative procedures improved the clinical attachment level and radiographic bone level in endodontic-periodontal lesions [20]. Hence, the periodontal regenerative approach should be considered in the treatment of combined EPL.

Another aspect of debate in EPL treatment is whether to graft the bony fenestration defects present or not. This is due to a consistent debate in the literature about the impact of the endodontic treatment on the healing potential of the periodontium [17–20]. Some authors have explained that this might be related to the size of bony defect and amount of bone destruction. Therefore, when increased periodontal tissue destruction occurs, an additional second procedure of bone grating can be beneficial to improve periodontal status and periodontal healing around the involved tooth, thus improving overall tooth prognosis [21,22]. Several types of bone grafting materials have been used in regenerated such defects ranging from gold standard osteo-inductive grafts (e.g., autografts and allografts) to osteoconductive grafts (e.g., xenografts, alloplasts) [23,24].

The relationship between the pulpal and periodontal tissues has been extensively studied; however, the treatment protocols of managing combined EPL need more investigations. The remarkable potential of MTA to stimulate the biologic mechanisms necessary for repair and retention of involved teeth could contribute as an alternative filling material in combined EPL. Therefore, the aim of this in-vivo clinical study was to evaluate and compare the clinical periodontal parameters in patients with true combined EPL, treated with GP and MTA as an obturation material alone and with the addition of bone graft in such lesions. The null hypothesis was that the evaluation parameters will remain the same in all the patients with EPL treated with the four treatment modalities.

## 2. Materials and Methods

### *2.1. Ethical Approval, Study Population, and Design*

This randomized controlled double blinded trial was conducted between October 2017 to November 2018 in accordance with the Helsinki Declaration of 1975, as revised in 2013. The protocol was approved by the Institutional Committee of Research Ethics at the King Saud University, Riyadh, Saudi Arabia (No. E-18-3540) and registered at ClinicalTrials.gov (NCT04274498).

Each participating patient signed an informed consent form after the details of the study had been explained to them. In addition, each patient was informed that they could withdraw from the study at any time without jeopardizing their rights to proceed with dental care at the Dental University Hospital.

The study was performed at the Dental University Hospital of King Saud University, Riyadh Saudi Arabia. 25–55 year-old Saudi patients diagnosed to have an upper anterior non-vital single rooted tooth with true combined EPL were recruited for this study. All subjects were medically fit and had no history of orthodontic, periodontal, or prosthodontic treatment. Exclusion criteria were as follows: (1) uncontrolled systemic disease or condition that alters bone metabolism (i.e., diabetes, osteoporosis, osteopenia, hyperparathyroidism, or Paget's disease); (2) history of oral cancer, sepsis, or adverse outcomes to oral procedures; (3) long term use of antibiotics (> 2 weeks in the past two months); and (4) use of medications known to modify bone metabolism (i.e., bisphosphonates, corticosteroids). Upper anterior single rooted teeth with true combined EPL confirmed by Cone beam computed tomography radiograph (CBCT) using a limited field of view (FOV) at a 0.200-mm voxel size, 96 kV, and 11.0 mA with an exposure time of 12s using a Planmeca scanner (Planmeca ProMax; Planmeca, Helsinki, Finland) were included in the study. Teeth with fractures, external or internal resorption or associated with pathological cysts were excluded.

The sample size was determined by G Power software where a confidence level was set at 95%, a power level of 80%, and a moderate effect size from the total sample size was calculated to be of 120 teeth ($n$ = 120), randomly assigned to each of study's four groups ($n$ = 30/group). For randomization, a locked computer software (Minitab 1.5, Minitab, State College, PA, USA) was used to allocate patients to one of the proposed study's four groups. Blocked randomization (block size = 3) was performed to maintain equal group size. The allocation result was concealed in a closed envelope and disclosed to endodontist and periodontist only on the day of the appointment.

The details of the study groups and the materials used in the study are presented in Table 1.

**Table 1.** Details of the tested groups and materials used in the research study.

| S. No. | Group ($n$ = 120) | Treatment | Obturation/Graft Material | Trade Name | Lot Number |
|---|---|---|---|---|---|
| 1. | Group-I ($n$ = 30) | Conventional non-surgical RCT performing standard methodology | Gutta Percha | Obtura Gutta Percha Bar PK/100 Bar | FD-MT057 |
| 2. | Group-II ($n$ = 30) | Conventional non-surgical RCT performing standard methodology | Mineral trioxide aggregate | ProRoot MTA, White, 10 × 0.5 g | MP-A040500000400 |
| 3. | Group-III ($n$ = 30) | Conventional non-surgical RCT performing standard methodology | Gutta Percha | Obtura Gutta Percha Bar PK/100 Bar | FD-MT057 |
| | | + Bone grafting | Bone graft | AlloOss. Natural Blend Cortico/Cancellous Particulate, 500–1000 mic. | SKU: 01-108-201 |
| 4. | Group-IV ($n$ = 30) | Conventional non-surgical RCT performing standard methodology | Mineral trioxide aggregate | ProRoot MTA, White, 10 × 0.5 g | MP-A040500000400 |
| | | + Bone grafting | Bone graft | AlloOss. Natural Blend Cortico/Cancellous Particulate, 500–1000 mic. | SKU: 01-108-201 |

### *2.2. Screening and Treatment Visits' Sequence:*

The first dental visit was performed for examination and assigning cases to each study group. The second dental visit included scaling and root plaining around the tooth and RCT performed by a blinded endodontist. The third dental visit was scheduled one week after RCT was completed also with a blinded periodontist to perform periodontal surgery for group III and IV.

### *2.3. Endodontic Treatment*

Each patient received local anesthesia of 2% lidocaine with 1:100,000 adrenaline (DENTSPLY Pharmaceutical, York, PA, USA). A rubber dam was placed, and the endodontic access opening was modified using Endo Access bur no. A0164 (Dentsply Maillefer, Ballaigues, Switzerland) and slow speed diamond KGS3203 (Dentsply Maillefer, Ballaigues, Switzerland). Filing was completed to the appropriated determined working length by an electronic apical locator (Root ZX, J. Morita Corp., Osaka, Japan) and peri-apical radiographs. All root canals were instrumented by standardized apical-coronal preparation techniques and were prepared with hand K-files at the established working lengths. Sizes 3, 4, or 5 Reamer T (Pierce Co., Tokyo, Japan) were used to flare the canal orifice of the roots. All apical preparations were enlarged to three sizes greater than the initial size of the file bound at the full working length. Root-canal irrigation with a combination of 2.5% sodium hypochlorite was performed during root canal preparation by a root-canal syringe. After preparation the patency of the apical foramen was confirmed by inserting a #15 K-file 1 mm through the working length. The root canal was irrigated and then dried with paper points (Absorbent paper points; Zippere, West Palm Beach, FL, USA). For GP group; The Obtura II system was prepared according to the manufacturer's instructions (Obtura II Operator's Manual 1993). Silver injection needles of 20 and 23 gauge were used for all obturations and a silicone stop was placed 2 to 5 mm from the working length. Root-canal sealer (Canals N, Showa Dental Co., Tokyo, Japan) was placed into the canal using a paper point. At the time of obturation, injection of the thermo plasticized GP was performed twice, separately. First, the needle was inserted in the apical direction until it bound to the canal wall, and the thermo plasticized GP heated to 200°C in the delivery system was injected. The needle was removed after injecting a few millimeters of GP near the tip of the preparation. The softened GP in the apical portion was then vertically condensed to the apex with a hand plugger dipped in alcohol to avoid adherence to the GP. The remaining root-canal space was then back-filled in increments until GP was observed in the cervical aspect of the root.

As for the MTA treatment group; MTA (ProRoot, Dentsply/Tulsa Dental, Tulsa, OK, USA) was mixed according to the manufacturer's instructions and was compacted in the root canal system. MTA was initially carried to the apical third using an MTA carrier and compacted using wet cotton rolled on a sterile reamer. The entire root canal system was filled with MTA. Furthermore, a squeezed dry cotton pellet was kept in the chamber and the access cavity was sealed with a temporary restorative material (Cavit-G, 3M ESPE, St. Paul, MN, USA). After 48 hours, the temporary restoration was removed and the cavity was restored with a glass ionomer base (Ketac Molar Easymix 3M ESPE) and light cure composite resin (Z 350, 3M ESPE). All RCT treatments were performed by one endodontist (A.Q).

### *2.4. Bone Grafting Surgical Procedure*

Local anesthesia of 2% lidocaine with 1:50,000 adrenaline (DENTSPLY Pharmaceutical, York, PA) was applied. Sulcular incision extended to the labial surface of one adjacent tooth on each side and two vertical incisions to enhance visibility and accessibility, full thickness mucoperiosteal flap was reflected on buccal side, and all granulation tissue was removed from buccal bone defect using a surgical curette*. Furthermore, the defect's dimensions were measured using a periodontal probe*.

Afterwards the defect was packed with DFDBA (cortico-cancellous) (ACE Surgical Supply Co, Brockton, MA, USA) reaching the edges of adjacent bone level, then the grafted site was covered with a resorbable collagen membrane (ACE Surgical Supply Co, MA, USA) that was trimmed to cover the defect extending its edges 2 mm more than the defect dimensions. Flap was repositioned and sutured using a total of six simple interrupted sutures (vicryl restorable (3–0), ACE Surgical Supply Co, MA, USA). Patients returned 7–10 days post-surgery, sutures were removed and wound healing was evaluated. Overall uneventful healing was observed with no adverse events. (e.g., signs of inflammation and pus discharge, infection).

All periodontal surgeries were performed by the same periodontist (R.A).

### 2.5. Follow Up Visits

The follow up dental visit was scheduled 1, 3, 6 and 12 months post full treatment completion with assurance that all included teeth received permanent restoration. Clinical measurements including probing depth (PD) in mm, relative clinical attachment level (r CAL) which is expressed as the distance between cement–enamel junction (CEJ) to the depth at which the probe met resistance in mm, gingival recession which is the distance from the cemento–enamel junction to the depth of the free gingival margin, Keratinized Tissue Width (KTW), and gingival phenotype (G.Ph.) were recorded at baseline, and 3 and 6 and 12 months follow up visits.

### 2.6. Radiographic Evaluation

Radiographic parameters, including Cone Beam Computed Tomography) CBCT) and standardized vertical periapical radiograph, were taken and measured at baseline and 6 months later in order to follow the recommended after treatment protocol for all combined EPL, and The Cone Beam Computed Tomography Periapical Index (CBCTPAI) was used for evaluation of the healing. Success rate was observed and was defined as changes occurring toward level (0) (Intact periapical bone structures) or 1 (diameter of Periapical radiolucency > 0.5–1 mm) at 6 months and 1 year follow up periods [25].

## 5. Conclusions

A true combined endo-perio lesion may present with multiple pathogenesis ranging from simple to complex, rendering the outcome of the treatment unpredictable. The lesion from the root canal infection and periodontium must be treated with endodontic and periodontal treatment, respectively. Use of GP and MTA for root canal obturation along with periodontal therapy and bone augmentation helps in resolving the complex endo-perio lesions. Bone grafting in addition to obturation with MTA was found to be the best treatment strategy in management of EPL cases and is recommended for clinicians who are treating EPL patients. During the treatment planning for the EPL cases, an interdisciplinary approach is vital for the success/favorable outcomes of the treatment.

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**Institutional Review Board Statement:** The study was conducted according to the guidelines of the Declaration of Helsinki Declaration of 1975, as revised in 2013. The protocol was approved by the Institutional Committee of Research Ethics at the King Saud University, Riyadh, Saudi Arabia (No. E-18-3540) and registered at ClinicalTrials.gov (NCT04274498). This study was also approved by ethical committee at College of Dentistry research center (CDRC), King Saud University.

**Informed Consent Statement:** Informed consent was obtained from all subjects involved in the study.

**Data Availability Statement:** Data is available on request from corresponding author.

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## References

1. Verma, P.K.; Srivastava, R.; Gupta, K.; Srivastava, A. Combined endodontic-Periodontal lesion: A clinical dilemma. *J. Interdiscip. Dent.* **2011**, *1*, 119. [CrossRef]
2. Makeeva, M.K.; Daurova, F.Y.; Byakova, S.F.; Turkina, A.Y. Treatment of an Endo-Perio Lesion with Ozone Gas in a Patient with Aggressive Periodontitis: A Clinical Case Report and Literature Review. *Clin. Cosmet. Investig. Dent.* **2020**, *12*, 447–464. [CrossRef] [PubMed]
3. Oktawati, S.; Siswanto, H.; Mardiana, A.; Supiaty; Neormansyah, I.; Basir, I. Endodontic–periodontic lesion management: A systematic review. *Medicina Clínica Práctica* **2020**, *3*, 100098. [CrossRef]
4. Schmidt, J.C.; Walter, C.; Amato, M.; Weiger, R. Treatment of periodontal-endodontic lesions—A systematic review. *J. Clin. Periodontol.* **2014**, *41*, 779–790. [CrossRef] [PubMed]
5. Kuoch, P.; Bonte, E. Endoperiodontal Lesions and Chicago's New Classification of Periodontal and Peri-implant Diseases and Conditions. *J. Contemp. Dent. Pr.* **2020**, *21*, 798–802. [CrossRef]
6. Herrera, D.; Retamal-Valdes, B.; Alonso, B.; Feres, M. Acute periodontal lesions (periodontal abscesses and necrotizing periodontal diseases) and endo-periodontal lesions. *J. Periodontol.* **2018**, *89* (Suppl. 1), S85–S102. [CrossRef]
7. Jivoinovici, R.; Suciu, I.; Dimitriu, B.; Perlea, P.; Bartok, R.; Malita, M.; Ionescu, C. Endo-periodontal lesion—Endodontic approach. *J. Med. Life* **2014**, *7*, 542–544.
8. Rotstein, I.; Simon, J.H.S. Diagnosis, prognosis and decision-making in the treatment of combined periodontal-endodontic lesions. *Periodontol. 2000* **2004**, *34*, 165–203. [CrossRef] [PubMed]
9. Kambale, S.; Aspalli, N.; Munavalli, A.; Ajgaonkar, N.; Babannavar, R. A sequential approach in treatment of endo-perio lesion a case report. *J. Clin. Diagn. Res.* **2014**, *8*, ZD22–ZD24. [CrossRef]
10. Vishwanath, V.; Rao, H.M. Gutta-percha in endodontics-A comprehensive review of material science. *J. Conserv. Dent.* **2019**, *22*, 216–222. [CrossRef]
11. Abusrewil, S.M.; McLean, W.; Scott, J.A. The use of Bioceramics as root-end filling materials in periradicular surgery: A literature review. *Saudi Dent. J.* **2018**, *30*, 273–282. [CrossRef] [PubMed]
12. Tang, Y.; Li, X.; Yin, S. Outcomes of MTA as root-end filling in endodontic surgery: A systematic review. *Quintessence Int.* **2010**, *41*, 557–566.
13. Chong, B.S.; Ford, T.R.P.; Hudson, M.B. A prospective clinical study of Mineral Trioxide Aggregate and IRM when used as root-end filling materials in endodontic surgery. *Int. Endod. J.* **2003**, *36*, 520–526. [CrossRef]
14. Kim, S.; Song, M.; Shin, S.-J.; Kim, E. A Randomized Controlled Study of Mineral Trioxide Aggregate and Super Ethoxybenzoic Acid as Root-end Filling Materials in Endodontic Microsurgery: Long-term Outcomes. *J. Endod.* **2016**, *42*, 997–1002. [CrossRef]
15. Alsulaimani, R.S. Single-visit endodontic treatment of mature teeth with chronic apical abscesses using mineral trioxide aggregate cement: A randomized clinical trial. *BMC Oral Health* **2016**, *16*, 78. [CrossRef] [PubMed]
16. Dakó, T.; Lazăr, A.P.; Bică, C.I.; Lazăr, L. Endo-perio lesions: Diagnosis and interdisciplinary treatment options. *Acta Stomatol. Marisiensis J.* **2020**, *3*, 257–261. [CrossRef]
17. Sam, G.; Pillai, B.R. Evolution of Barrier Membranes in Periodontal Regeneration-"Are the third Generation Membranes really here?". *J. Clin. Diagn. Res.* **2014**, *8*, ZE14–ZE17. [CrossRef]

18. Stavropoulos, A.; Sculean, A. Current Status of Regenerative Periodontal Treatment. *Curr. Oral Health Rep.* **2017**, *4*, 34–43. [CrossRef]
19. Cortellini, P.; Stalpers, G.; Mollo, A.; Tonetti, M.S. Periodontal regeneration versus extraction and prosthetic replacement of teeth severely compromised by attachment loss to the apex: 5-year results of an ongoing randomized clinical trial. *J. Clin. Periodontol.* **2011**, *38*, 915–924. [CrossRef]
20. Oh, S.; Chung, S.H.; Han, J.-Y. Periodontal regenerative therapy in endo-periodontal lesions: A retrospective study over 5 years. *J. Periodontal Implant. Sci.* **2019**, *49*, 90–104. [CrossRef]
21. Alquthami, H.; Almalik, A.M.; Alzahrani, F.F.; Badawi, L. Successful Management of Teeth with Different Types of Endodontic-Periodontal Lesions. *Case Rep. Dent.* **2018**, *2018*, 1–7. [CrossRef]
22. Deepak, S. Surgical Management of Endo-Perio Lesion using Bonegraft and Guided Tissue Regeneration—A Case Report. *Int. J. Dent. Oral Sci.* **2020**, 19–23. [CrossRef]
23. Titsinides, S.; Agrogiannis, G.; Karatzas, T. Bone grafting materials in dentoalveolar reconstruction: A comprehensive review. *Jpn. Dent. Sci. Rev.* **2019**, *55*, 26–32. [CrossRef]
24. Kumar, P.; Vinitha, B.; Fathima, G. Bone grafts in dentistry. *J. Pharm. Bioallied Sci.* **2013**, *5* (Suppl. 1), S125–S127. [CrossRef]
25. Sanchez-Torres, A.; Sanchez-Garces, M.; Escoda, C.G. Materials and prognostic factors of bone regeneration in periapical surgery: A systematic review. *Med. Oral Patol. Oral Cir. Bucal* **2014**, *19*, e419–e425. [CrossRef] [PubMed]
26. Von Arx, T.; Britain, S.; Cochran, D.L.; Schenk, R.K.; Nummikoski, P.; Buser, D. Healing of periapical lesions with complete loss of the buccal bone plate: A histologic study in the canine mandible. *Int. J. Periodontics Restor. Dent.* **2003**, *23*, 157–167.
27. Estrela, C.; Bueno, M.R.; Azevedo, B.C.; Azevedo, J.R.; Pécora, J.D. A new periapical index based on cone beam computed tomography. *J. Endod.* **2008**, *34*, 1325–1331. [CrossRef] [PubMed]
28. Al Attas, M.A.; Edrees, H.Y.; Sammani, A.M.; Madarati, A.A. Multidisciplinary management of concomitant pulpal and periodontal lesion: A case report. *J. Taibah Univ. Med. Sci.* **2017**, *12*, 455–460. [CrossRef]
29. Raja, V.S.; Emmadi, P.; Namasivayam, A.; Thyegarajan, R.; Rajaraman, V. The periodontal-endodontic continuum: A review. *J. Conserv. Dent.* **2008**, *11*, 54–62. [CrossRef] [PubMed]
30. Abbott, P.V.; Salgado, J.C. Strategies for the endodontic management of concurrent endodontic and periodontal diseases. *Aust. Dent. J.* **2009**, *54*, S70–S85. [CrossRef]
31. Jayadev, M.; Shravani, G.S.; Karunakar, P.; Prasanna, J.S. Platelet-rich fibrin, "a faster healing aid" in the treatment of combined lesions: A report of two cases. *J. Indian Soc. Periodontol.* **2014**, *18*, 651–655. [CrossRef]
32. Gomes, B.P.F.D.A.; Herrera, D.R. Etiologic role of root canal infection in apical periodontitis and its relationship with clinical symptomatology. *Braz. Oral Res.* **2018**, *32*, e69. [CrossRef] [PubMed]
33. Shetty, K.; Habib, V.A.; Shetty, S.V.; Khed, J.N.; Prabhu, V.D. An assessment of coronal leakage of permanent filling materials in endodontically treated teeth: An in vitro study. *J. Pharm. Bioallied Sci.* **2015**, *7*, S607–S611. [CrossRef] [PubMed]
34. Girish, K.; Mandava, J.; Chandra, R.R.; Ravikumar, K.; Anwarullah, A.; Athaluri, M. Effect of obturating materials on fracture resistance of simulated immature teeth. *J. Conserv. Dent.* **2017**, *20*, 115–119. [CrossRef] [PubMed]
35. Olczak, K.; Pawlicka, H. Mineral trioxide aggregate in treatment of permanent teeth with open apex and endo-perio lesions. A case report. *Eur. J. Paediatr. Dent.* **2015**, *16*, 287–289.
36. Okiji, T.; Yoshiba, K. Reparative dentinogenesis induced by mineral trioxide aggregate: A review from the biological and physicochemical points of view. *Int. J. Dent.* **2009**, *2009*, 1–12. [CrossRef] [PubMed]